

Advanced SQL: Functions

Q1: Employees table, show all employees with their salary.
If salary is NULL, display 0

Answer

```
SELECT employee_id,
       name,
       IFNULL(salary, 0) AS salary-with-default
FROM employee;
```

Employee-id	Name	Salary-with-default
1	Ahu	5000
2	Bob	0
3	Charlie	7000
4	Dana	0

Q2: Show all names with their department. If department is NULL,
Not assigned

```
SELECT employee_id,
       name,
       department_name
```

~~FROM~~ employees

WHERE COALESCE(department, 'Not Assigned') AS department
FROM employees;

Table: Employees

employee_id	name	department_name
1	Alice	HR
2	Bob	IT
3	Charlie	Not Assigned
4	Dana	Finance.

Q3. Find orders with NULL customer_id using ISNULL()

```
SELECT order_id,
       customer_id
```

```
FROM orders
```

```
WHERE customer_id IS NULL;
```

Table: Orders

Order_id	Customer_id
103	NULL

Q4. Show all orders, if delivery-date is NULL, show 'Pending'

```
SELECT Order_id, customer_id, delivery_status
       COALESCE(CAST(delivery_date AS VARCHAR), 'Pending') AS delivery_date
FROM orders;
```

Order_id	Customer_id	delivery_status
101	201	2024-12-01
102	202	Pending
103	NULL	2024-12-03

Q5. Show all student and their grades. Replace NULL with 0
 SELECT student_id, name, COALESCE (grade, 0) AS final_grade
 FROM Students;

Student_id	Name	Final_grade
1	Ethan	85
2	Marya	0
3	Olivia	90

Q6. Count Students who haven't been graded.

SELECT
 COUNT (*) AS not_graded_count
 FROM Students
 WHERE grade is NULL;

not_graded_count
1

Q7. Show product name, price, and final price after discount (assume 0 if discount is NULL).

SELECT product_id, name, final_price
 Price - COALESCE (discount, 0) AS final_price
 FROM Products

Product_id	Name	Final-price
501	Keyboard	25
502	Mouse	10
503	Monitor	100

Q8 Count how many have no email

```
SELECT
  customer_id,
  name,
  COALESCE(email, 'not provided') AS email_status
FROM customers;
```

```
SELECT COUNT (*) AS missing_email_count
FROM customers;
Where email IS NULL
```

Missing_email_count
2

Q9. Show all customers with email. IF NULL, display "No Email"

```
SELECT customer_id,
  name,
  COALESCE(email, 'no email') AS email_display
FROM customers;
```

Customer_id	Name	Email_display
1.	Linda	No Email
2.	Joseph	Joseph@email.com
3.	Nia	No Email-

Q10. Show payment details with method replaced by 'Unknown' if null
 SELECT payment-id,
 COALESCE (Method, 'Unknown') AS Method_display,
 Status
 FROM payment;

Payment-id	Method-display	Status
301	Credit	NULL
302	Paypal	Success
303	Unknown	Failed-

Q11. Show items and their quantity (0 if null)
 SELECT Item-id,
 Item-name,
 COALESCE (quantity, 0) AS quantity-checked
 FROM Inventory;

Item-id	Item-name	quantity-checked
1	Pen	0
2	Notebook	150
3	Eraser	0

Q12. Show employee ID and the first available value among bonus or commission.

SELECT employee_id,
COALESCE (bonus, commission) As first_available_reward
FROM Employees_Ext;

emp-id	First-available_reward
1	300
2	100
3	NULL

Q13. Count desks that don't have a room assigned

no-room_count
2

Q14. Show attendance records with status. Replace NULL with 'Not Marked'

SELECT student_id,
date
COALESCE (status, 'not specified') As attendance_status
FROM Attendance

Student_id	date	attendance_status
1	2025-04-01	Not marked
2	2025-04-10	Present

Q15. Show amount ID, amount_type (or 'unknown') and balance (or 0)

```
SELECT amount_id,
COALESCE (amount_type, 'unknown') AS type_display,
COALESCE (balance, 0) AS balance_display
FROM Bank_accounts
```

Table: Bank-Account

amount_id	type_display	balance_display
A1	Savings	0
A2	Current	5000
A3	Unknown	2000

Q16. Show all projects with a start date. IF start_date IS NULL, display 'TBD'

```
SELECT project_id,
title,
COALESCE (start_date, 'TBD') AS start_display
FROM Projects;
```

Table 12: Projects

Project_id	Title	Start_display
1	Website Revamp	2025-01-01
2	Mobile App	TBD
3	Data Migration	TBD

Question 17: Display reviews showing comment (or 'No Comment') and rating (or 0)

```
SELECT review_id,
       product_id,
       COALESCE (comment, 'No Comment') AS comment_display,
       COALESCE (rating, 0) AS rating_display
FROM Reviews;
```

Table 13: Reviews

review_id	product_id	comment_display	rating_display
1	501	Great product	4
2	502	No comment	0
3	503	Works fine	3

Question 18: Show the supplier contact number. Use COALESCE (phone, alt_phone, 'No contact')

```
SELECT supplier_id,
       name,
       COALESCE (phone, alt_phone, 'No contact') AS
       contact_number
FROM Suppliers;
```

Table 14: Suppliers

Supplier-id	name	Contact-number
1	Global Goods	123456789
2	Best Supplies	987654321
3	ValueSource	No Contact

Question 19. Show all users and their preferences, Replace all NULLS with defaults.

```
SELECT user_id,
       COALESCE(theme, 'Light') AS theme_set,
       COALESCE(language, 'English') AS language_set,
       COALESCE(timezone, 'UTC') AS timezone_set
FROM user_settings;
```

Table: User-Settings

user_id	theme_set	language_set	timezone_set
1	Light	English	UTC
2	Dark	English	UTC+1
3	Light	English	UTC

Question 20. Show maintenance log with:

```
SELECT record_id,
       machine_id,
       COALESCE(issue, 'Unknown issue') AS issue_log,
       COALESCE(technician, 'Not Assigned') AS technician_name
FROM Maintenance;
```

Table: Maintenance

record_id	machine_id	issue_log	technician_name
1	M101	Overheating	Not Assigned
2	M102	Unknown issue	Not Assigned
3	M103	Jammed	Alex